



Rational design and modular synthesis of biodegradable ionizable lipids via the Passerini reaction for mRNA delivery

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Affiliations are included on p. 11.

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The ionizable lipid component of lipid nanoparticle (LNP) formulations is essential for mRNA delivery by facilitating endosomal escape. Conventionally, these lipids are synthesized through complex, multistep chemical processes that are both time-consuming and require significant engineering. Furthermore, the development of new ionizable lipids is hindered by a limited understanding of the structure-activity relationships essential for effective mRNA delivery. In this work, we have developed a modular platform utilizing the Passerini reaction to rapidly generate large, chemically diverse libraries of biodegradable ionizable lipids. This high-throughput approach enables the systematic exploration of various lipid components—head groups, tails, and spacers—and their impacts on mRNA delivery efficiency. By investigating the hydrogen bonding potential between the lipid's head groups and the mRNA's ribose phosphate complex, we found that optimizing the methylene units between the lipid's head groups and linkages could enhance endosomal escape and, consequently, mRNA delivery efficiencies. Leveraging this insight, our platform has led to the identification of the biodegradable ionizable lipid A4B4-S3, which outperforms the current clinical benchmark, SM-102, in gene editing efficacy in mouse liver following systemic administration and demonstrates the promise for repeat-dose protein replacement treatments. This work not only offers a rapid, scalable method for ionizable lipid synthesis but also deepens our understanding of their structure-activity relationships, paving the way for more effective mRNA therapeutics.

lipid nanoparticles | ionizable lipid | mRNA delivery | gene editing

mRNA-containing lipid nanoparticles (mRNA-LNPs) are a powerful nanomedicine platform, as demonstrated by the crucial role mRNA-LNP vaccines have played during the coronavirus disease 2019 (COVID-19) pandemic (1). This platform is of great interest as a tool for vaccination, protein replacement therapy, and genome editing (2); however, achieving efficient delivery of mRNA remains one of the key challenges in fully realizing the therapeutic potential of this technology (3). Classic LNP formulations consist of four main components: an ionizable lipid, a phospholipid, cholesterol, and a polyethylene glycol (PEG) conjugated lipid (4). For improvement of mRNA delivery, ionizable lipids have garnered significant attention due to their diverse chemical structures and their critical roles in stabilizing the mRNA cargo and facilitating endosomal escape (5). All three Food and Drug Administration approved RNA-LNP drugs (Onpattro, Comirnaty, and Spikevax) feature distinct ionizable lipids crucial for their performance (6). Despite advancements in developing new ionizable lipids that improve mRNA loading and delivery (7), significant challenges persist in designing new lipid structures (8). Traditional ionizable lipid synthesis is marked by a series of complex steps, including chemical protection and deprotection, catalyst use and removal, solvent exchange, and complicated purification processes (9). These time-consuming procedures necessitate customization for each synthetic reaction, limiting throughput and constraining the exploration of new lipid structures (10).

Combinatorial chemistry is particularly valuable in the development of ionizable lipids by facilitating the creation of extensive libraries through systematic combinations of multiple building blocks (9). This approach allows researchers to quickly and efficiently generate diverse libraries of ionizable lipid structures, which can then be screened for their capacity to encapsulate and effectively deliver mRNA to target cells. Through this high-throughput approach, the potential for discovering more efficient and less toxic ionizable lipids is vastly increased. Moreover, combinatorial chemistry enhances the systematic exploration of structure-activity relationships (SAR) within lipid libraries. This method allows for the simultaneous examination of multiple structural variants, enabling the identification of critical features that impact lipid functionality and inform the design of effective mRNA delivery systems. Numerous reaction types including aza-Michael

Significance

Effective delivery strategies are vital for message RNA (mRNA) vaccines and therapeutics, with lipid nanoparticles (LNPs) being forefront. Developing ionizable lipids, a key LNP component, traditionally involves complex, multistep synthesis and is hindered by limited knowledge of their structure-activity relationships for mRNA delivery. In this study, we introduce a modular platform that facilitates the rapid creation of extensive, chemically diverse libraries of biodegradable ionizable lipids. This platform led to the identification of A4B4-S3, which is formulated in mRNA-loaded LNPs, and outperforms the clinical benchmark SM-102 in mouse liver gene editing. Our results highlight the critical role of hydrogen bonding between ionizable lipids and ribose-phosphate complexes in mRNA delivery efficiency, offering a high-throughput approach for lipid synthesis and insights into their design.

Competing interest statement: Y.X. and B.L. have filed a patent for the development of the described lipids.

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addition of amines to acrylamides or acrylates, ring-opening of epoxides with amines, reductive amination of aldehydes, and thio-ene click reactions (8, 11–13) have been utilized to design and create new ionizable lipids capable of delivering mRNA to various tissues for diverse biomedical applications, such as cancer vaccines, retina tissue delivery, and lung gene editing (14, 15). However, these reactions often face limitations in the structural diversity of the ionizable lipids they can produce due to their two-dimensional nature.

To address this limitation, combinatorial chemistry, utilizing multicomponent reactions (MCR), has emerged as an effective strategy for the high-throughput synthesis (HTS) of ionizable lipids. Notable examples include the Ugi-based three-component reaction (3CR), which can rapidly generate a combinatorial library of ionizable lipids within 24 h at room temperature, thereby enhancing both the speed and diversity of lipid production due to its three-dimensional capability (6, 16, 17). However, integrating biodegradable bonds into lipid structures requires the customized synthesis of biodegradable lipid tails as building blocks for the Ugi-based 3CR, adding significant time and effort. Additionally, this approach often results in inconsistent and low yields, complicating high-throughput screening processes. While a recent work improved the yield of this platform through the incorporation of a nontoxic catalyst (6), there remains a critical need for a facile, catalyst-free MCR platform capable of rapidly synthesizing biodegradable and diverse lipids with high yields (18, 19). Identifying suitable MCRs beyond the Ugi reaction that fulfill the complex requirements for constructing ionizable lipid structures continues to be a significant challenge in combinatorial chemistry.

Here, we introduce a modular system for synthesizing ionizable lipids utilizing the catalyst-free Passerini-three components reaction (P-3CR) method. This approach enables the combinatorial creation of ionizable lipids under mild conditions, achieving high yield, diversity, efficiency, and biodegradability. The system is designed around three key components: an ionizable amine-containing head group, and two tails, A and B (20). Utilizing this method, we synthesized 144 chemically diverse ionizable lipids featuring nine distinct headgroups with linear, cyclic, and aromatic structures, along with eight different tails varying in length and saturation. Each lipid produced features an α -acyl amide bond, enhancing physiological biodegradability and reducing potential adverse reactions (21, 22) making them suitable for multidose regimens (23). Moreover, our results demonstrate that variations in the methylene units between the head groups and linkages of the ionizable lipids alter hydrogen bonding interactions with mRNA ribose phosphate complexes. This interaction, especially significant with guanine (G) and cytosine (C) bases, crucially enhances the efficiency of LNP-mediated mRNA delivery. Driven by this finding, we identified an ionizable lipid, A4B4-S3, which significantly improves gene editing efficiency in mouse liver following systemic administration, surpassing the performance of the clinically benchmarked lipid SM-102. Additionally, this biodegradable lipid also shows promise for repeated-dose protein replacement therapy. Overall, this platform not only streamlines the synthesis of ionizable lipids but also enriches our understanding of their design principles and structure-activity relationships, crucial for enhancing mRNA delivery.

Results and Discussion

P-3CR for Ionizable Lipid Library Synthesis. To synthesize our ionizable lipid library, we used the P-3CR. This one-pot reaction combines carboxylic acids (representing tertiary amine-containing

headgroups), isocyanides (representing Tail A), and aldehydes (representing Tail B) (Fig. 1A). The P-3CR reaction enables efficient and versatile synthesis and purification processes for ionizable lipids, reducing the number of synthesis and purification steps, omitting the use of catalysts or couplers, only requiring mild conditions, such as room temperature, a brief reaction time, and neutral pH levels and exhibiting excellent functional group tolerance (24). Newer generations of ionizable lipids are often crafted with ester bonds for rapid elimination, aiming to improve tolerability during payload delivery. Enriching ionizable lipids with ester bonds offers robust stability at physiological pH while conferring the ability to undergo enzymatic hydrolysis in cells and tissues. Enhanced metabolism and degradation of ionizable lipids *in vivo* are particularly important for therapeutic applications that require repeated administration. The P-3CR reaction involves the condensation of isocyanide, aldehyde, and carboxylic acid groups, succeeded by a Mumm rearrangement, ultimately producing the α -acyl amide scaffold. These α -acyl amide scaffolds feature a hydrolyzable ester bond, conferring the advantages of biodegradability while eliminating the necessity for additional esterification reactions, underscoring the method's efficiency and simplicity compared to alternative multicomponent reactions (10).

The efficiency and versatility of the P-3CR reaction make it conducive to HTS. Using the synthetic methodology of P-3CR, we constructed an initial combinatorial library of 144 ionizable lipids (Fig. 1B and *SI Appendix, Table S1*). Guided by established principles of ionizable lipid design, the library consisted of i) diverse carboxylic acid headgroups including linear alkane-based, cycloalkane-based, and aromatic ring-based structures within tertiary amine backbones; ii) alkyl tails with lengths ranging from C10 to C18; iii) tail structures featuring branching, unsaturation, and additional ester bonds. Conveniently, the classic Passerini reaction generates lipids at an acceptable purity for high-throughput screening (Fig. 1C) (25, 26). The resulting ionizable lipids were formulated into LNPs using a classical four-composition formulation ratio established by Moderna for SM-102, containing 1,2-distearoyl-sn-glycero-3-phosphocholine (DSPC) as a helper lipid, cholesterol, 1,2-dimyristoyl-sn-glycerol-3-phosphoethanolamine-N-[methoxy-(polyethyleneglycol)-2000] (C14-PEG), and the synthesized ionizable lipids from the P-3CR library (Fig. 1D) (27).

Optimization of Iso-A11B5C1 LNP for Muscle-Specific mRNA Delivery. To assess the ability of the synthesized lipids to deliver mRNA *in vivo*, the LNPs were formulated with firefly luciferase mRNA (mLuc) using a microfluidic T-junction device for *in vivo* batch testing and physical characterization (Fig. 2A) (28). While the P-3CR facilitated rapid synthesis of a large, diverse array of ionizable lipids, a batch-based testing approach was employed to efficiently assess all 144 LNPs *in vivo* (16). Previous research indicates a significant correlation between the tertiary amine-containing headgroups of ionizable lipids and their mRNA transfection potency (29); thus, the 144 LNPs were first categorized into nine groups (H1-H9) based on the headgroup structure of their ionizable lipid for subsequent characterization and batch testing (*SI Appendix, Figs. S1–S9*) (5, 30). The headgroups were further described as being linear (H1-H3), cyclic (H4-H6), and aromatic (H7-H9).

Before evaluating the transfection potency of mLuc-LNPs, their physicochemical properties were analyzed. The polydispersity index (PDI), size, and encapsulation efficiency (EE) analysis served as quality control for subsequent *in vivo* screening (*SI Appendix, Table S1*). There was no significant correlation between the physical properties of the LNPs and the chemical structure of the ionizable

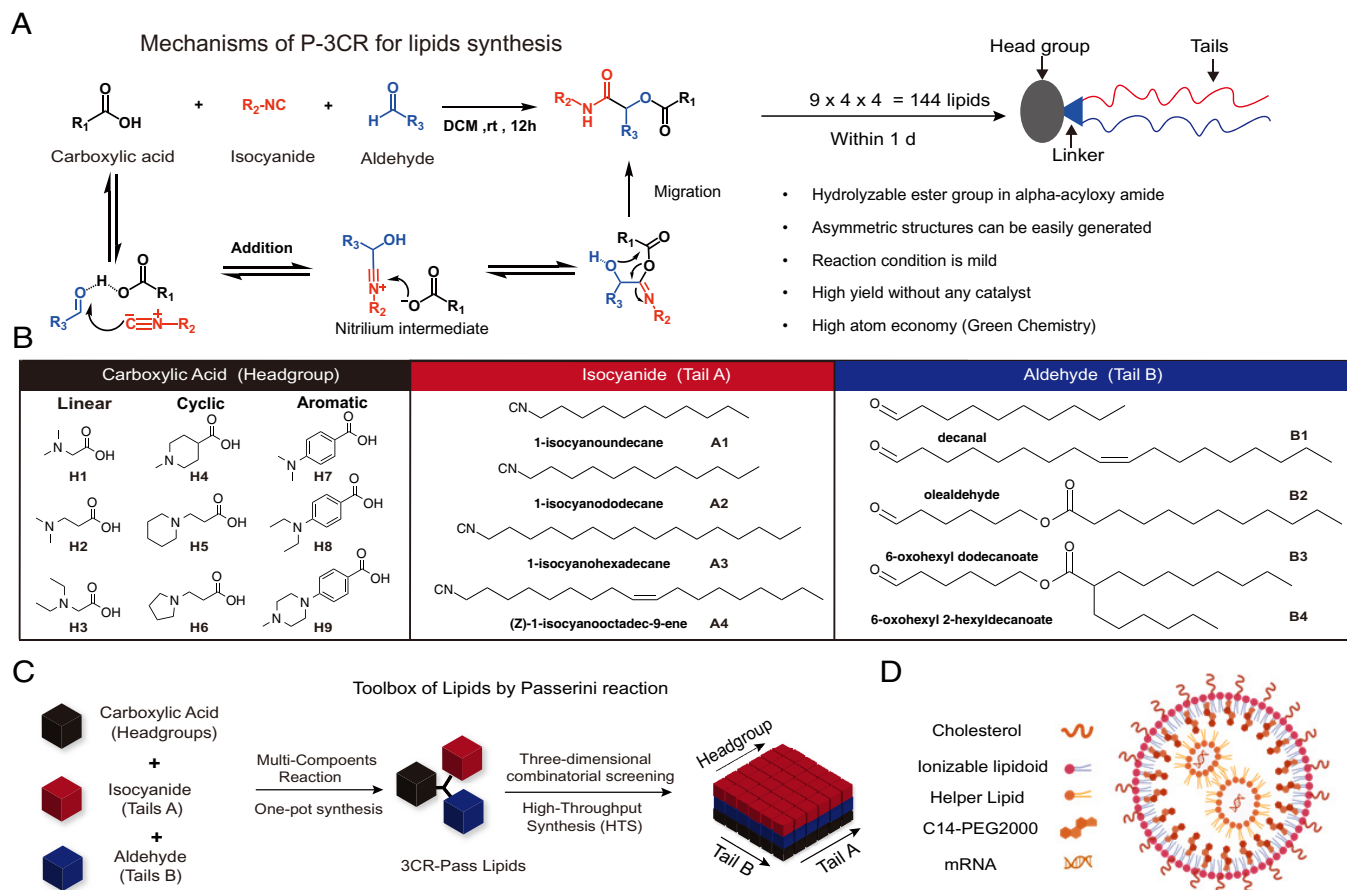


Fig. 1. Design and synthesis of ionizable lipids by Passerini reaction. (A) Schematic representation of the mechanism of P-3CR-based synthesis of ionizable lipids. (B) Elaborate structures of three constituents employed in a batch test for the synthesis of ionizable lipids in the P-3CR library, including carboxylic acid (headgroup), isocyanide (tail A), and aldehyde (tail B). (C) Schematic of the HTS of ionizable lipids based on P-3CR. (D) Classical formulations for the preparation of mRNA-LNPs.

lipid headgroups (Fig. 2B). For an initial assessment of the delivery efficiency of the 144 ionizable lipids, LNPs containing ionizable lipids that shared the same headgroup were pooled and administered via intramuscular (I.M.) injection (Fig. 2C). Notably, linear and cyclic alkane headgroups achieved the highest transfection potency, while aromatic headgroups showed significantly weaker transfection potency. Dimethylglycine (H1), 3-(dimethyl amino) propionic acid (H2), and 3-(1-pyrrolidinyl) propanoic acid (H6) lipids outperformed the other headgroups, with H2 achieving the highest mLuc expression (Fig. 2D). Thus, ionizable lipids containing H2 were advanced to the next stage of batch testing wherein H2 LNPs were pooled based on shared Tail A structure and further assessed after I.M. administration. The results showed that (Z)-1-isocyanooctadec-9-ene (A4) containing unsaturated bonds achieved the highest transfection (Fig. 2E). In the third round of testing, all four LNPs with ionizable lipids containing headgroup H2 and tail A4 were tested separately in mice (Fig. 2F–H). The top two lipid candidates, H2A4B3 and H2A4B4, were assessed in a head-to-head comparison with the FDA-approved ionizable lipid MC3. Interestingly, despite lipids H2A4B3 and H2A4B4 differing only in one tail chain, the LNP containing H2A4B4 significantly outperformed both H2A4B3 and MC3 (Fig. 2I–K). To further compare the effects of linear and bifurcated tails on mRNA transfection efficiency, we also synthesized H2A4-Ole and H2A4-Butyl based on the previous structures. The results showed that ionizable lipids with branched tails exhibited significantly better mRNA transfection efficiency than those with linear tails. A possible explanation is that branched-tail ionizable

lipids are more conducive to forming cone-shaped ion pairs with anionic lipids. This formation drives the transition from a bilayer structure to the inverted hexagonal H_{II} phase, promoting membrane fusion/disruption, endosomal escape, and cargo release into the cytosol (31).

H2A4B4, the top-performing lipid in this library, contains three substructures: i) a nonaromatic amine head group, ii) a hydrophobic branched tail, and iii) an unsaturated hydrophobic linear tail. The electrostatic interaction between the ionizable head group and the negatively charged RNA backbone is a crucial step in the formation of RNA complexes. Alkyl headgroups in ionizable lipids exhibit lower planar rigidity compared to aromatic headgroups, potentially boosting their interaction with the RNA ribose-phosphate complex and enabling stable RNA complex formation. The structure of the hydrophobic tail influences various factors such as the pK_a, lipophilicity, and transition temperature of the lipid (32, 33). Optimal hydrophobic chain length and unsaturation facilitate lipid phase transitions, regulating LNP mobility and membrane fusion with endosomes. Incorporating unsaturated and branched fatty acids into lipid tails within this library led to heightened delivery efficiencies, likely attributable to their lower transition temperatures and ability to enhance membrane fluidity. These findings also indicate that combining a linear and a branched tail yields the highest mRNA transfection potency (Fig. 2I); this asymmetric structural characteristic bears resemblance to SM-102, the ionizable lipid utilized in Moderna's COVID-19 vaccine (34).

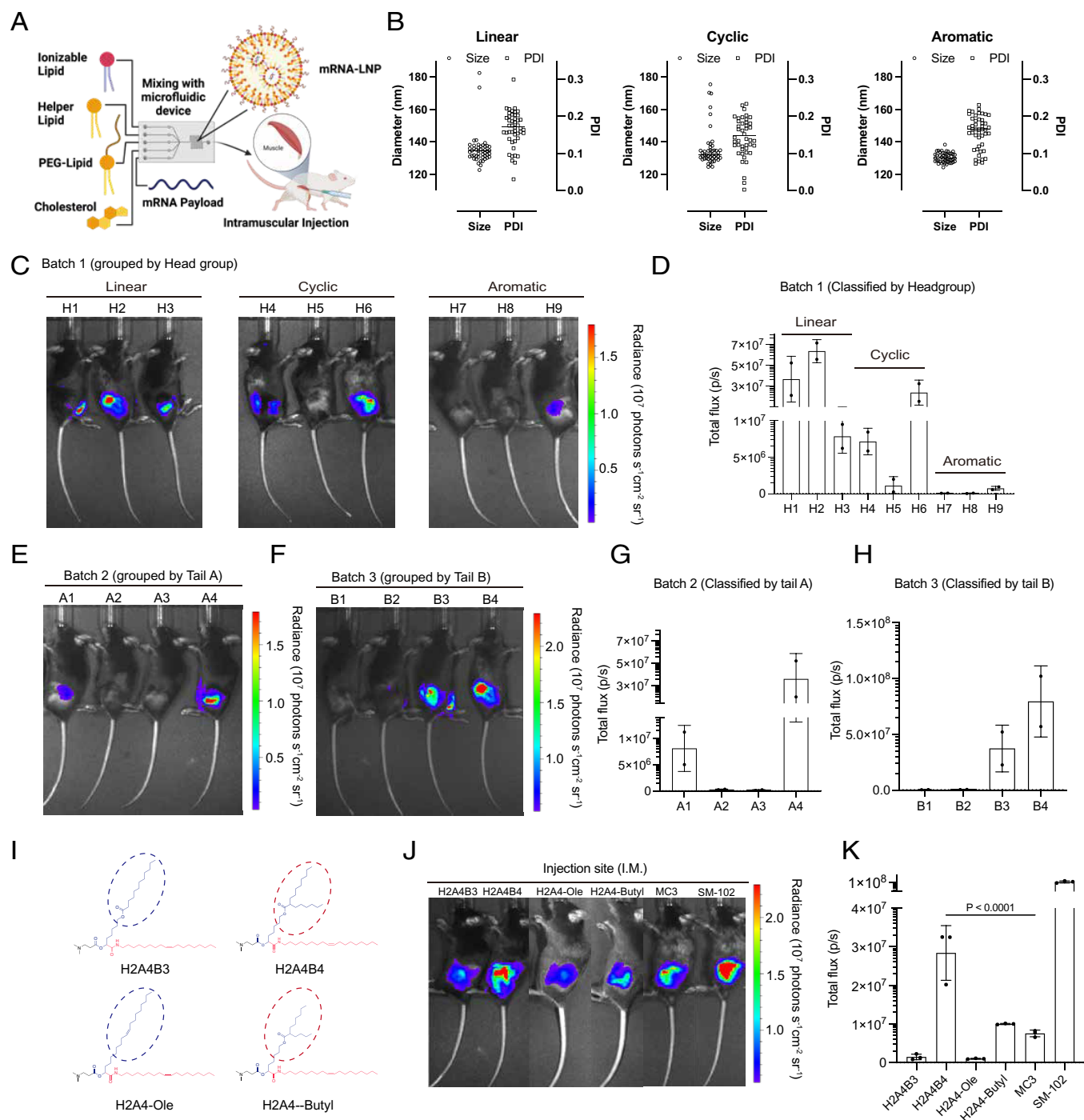


Fig. 2. Batch screening of lipids for Fluc mRNA (mLuc) delivery. (A) Schematic of in vivo orthogonal batch test of LNPs from P-3CR library by I.M. injection. (B) The head groups were classified according to their chemical structure and the corresponding LNPs were characterized for their physicochemical properties. (C and D) Batch 1 analysis identified the best-performing head groups (0.32 mg kg^{-1} mLuc, 16 LNP mixtures per mouse, $n = 2$). (E) Batch 2 study identified one top-performing lipid tail (A4) (0.08 mg kg^{-1} mLuc, 4 LNP mixtures per mouse, $n = 2$). (F) Batch 3 determined the individual lipids with optimal transfection potency (0.05 mg kg^{-1} mLuc per mouse, $n = 2$). (G and H) Histogram analysis of batches 1 and 2. (I) The lipid structures were identified by batch test and lipids after the chemical modification (lipid H2A4B3, H2A4B4, H2A4-Ole, and H2A4-Butyl). (J and K) The final comparison with the benchmark MC3 and SM-102 (0.05 mg kg^{-1} mLuc per mouse, $n = 3$). Statistical significance was assessed using a two-tailed Student's *t* test. **P*-value < 0.05 , ***P*-value < 0.01 , ****P*-value < 0.005 . Data are presented as mean $\pm$ SEM.

The P-3CR Method Enables Modular Optimization of Ionizable Lipids for Increased mRNA Transfection Potency. To explore why, despite their relative structural similarity, lipid H2A4B4 achieved significantly higher mRNA delivery efficiency than H2A4B3, further investigation into their conformational relationship was performed. The structural difference between these lipids is the inclusion of a branched Tail B (H2A4B4) or a linear Tail B (H2A4B3) (Fig. 3A). Previous studies suggest

that branched tails may function as wedges, widening the gap between lipid-like molecules within the LNP bilayer (35). The resulting expanded space promotes amine protonation by mitigating the repulsion between neighboring tertiary amines, which is known to destabilize negatively charged endosomal membranes (2, 36). To further validate this hypothesis for why asymmetric branching chains exhibited superior mRNA transfection potency, the membrane fusion capability of H2A4B3

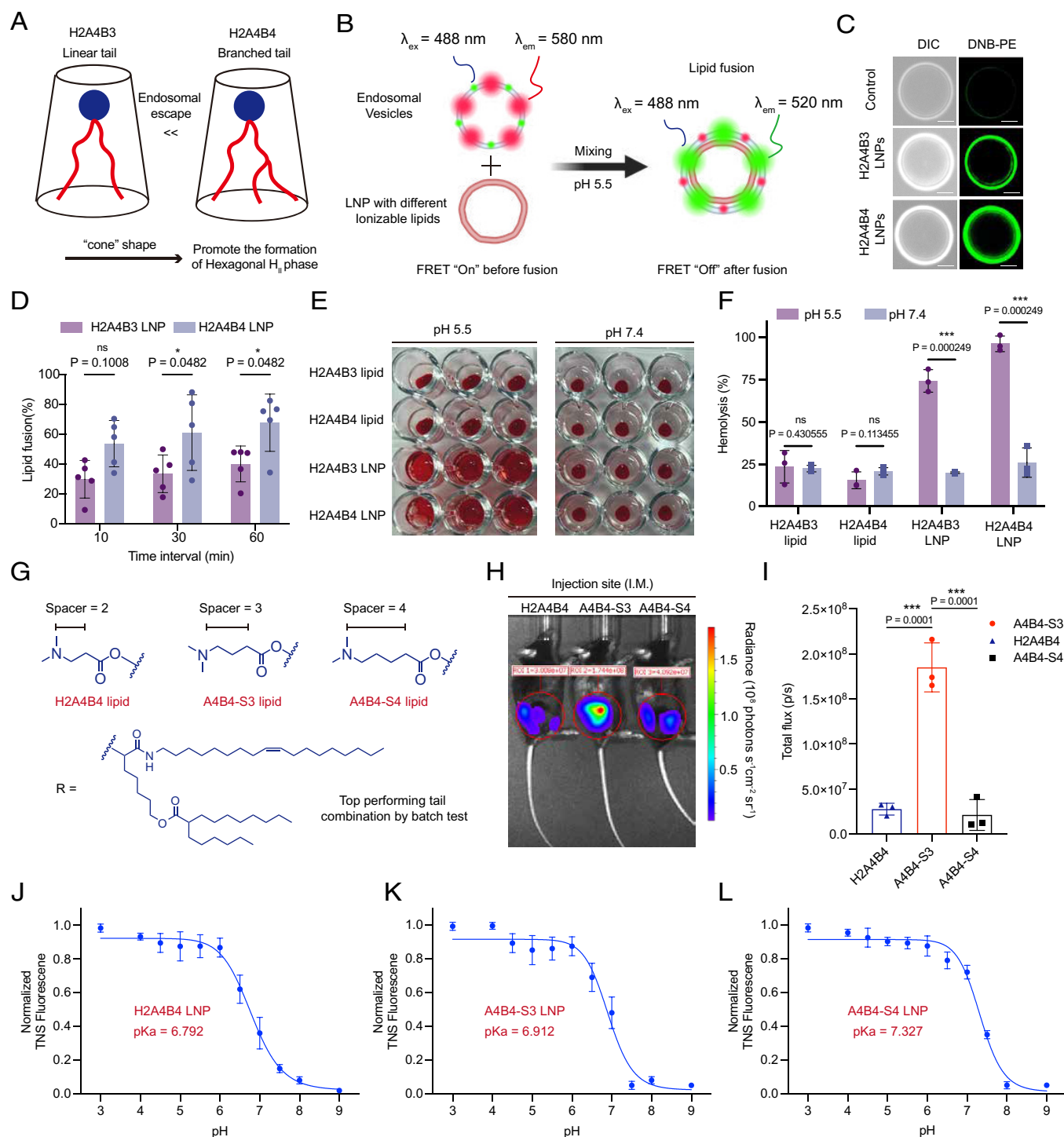


Fig. 3. Ionizable lipid-mediated mRNA delivery mechanism correlation with lipid tail structure in model membrane endosomal escape studies. (A) Ionizable lipid H2A4B4 tends to adopt a molecular “cone” shape, promoting the formation of inverted phases like the hexagonal H_{II} phase shown. (B) FRET assay to assess lipid membrane fusion (incubated with endosomal mimic at pH 5.5 in 37 °C for 60 min). (C) Representative fluorescent images after the lipid dissociation (DNB-PE: λ_{ex} = 488 nm, λ_{em} = 520 nm). (D) Histogram analysis of lipid fusion efficacy in different time intervals (10, 30, and 60 min, n = 5). (E) Hemolysis analysis of variable lipid and LNP in acidic and neutral pH conditions (n = 3). (F) Histogram analysis of hemolysis percentage in different pH conditions compared to the control group (n = 3). (G) Lipid structure with the top performing tail combination containing different spacers (number of methylene units = 2, 3, and 4). (H) LNP formulated with mLuc were injected intramuscularly into mice (0.05 mg kg^{-1} mLuc per mouse, n = 3). (I) The quantification data reveal that among the three LNPs, A4B4-S3 LNP stands out as the most efficient delivery vehicle. Data are shown as mean $\pm$ SEM (n = 3). (J–L) The TNS fluorescence was normalized to assess the apparent pK_a of various LNPs. Statistical significance was assessed using a two-tailed Student's t test. * P -value < 0.05, ** P -value < 0.01, *** P -value < 0.005. Data are presented as mean $\pm$ SEM.

and H2A4B4 LNPs was evaluated through fluorescence resonance energy transfer (FRET), guided by the mechanism of ionizable lipid-mediated endosomal escape (Fig. 3B) (37). Two DOPE-conjugated FRET probes, N-(7-Nitrobenz-2-Oxa-1,3-Diazol-4-yl)-1,2-Dihexadecanoyl-sn-Glycero-3-Phosphoethanolamine

(NBD-PE) and lissamine rhodamine B sulfonyl (Liss Rho-PE) were employed in an endosome-like structure where the proximity of the lipids converts NBD fluorescence to rhodamine, resulting in muted NBD fluorescence overall. Upon fusion with LNPs, the separation between the two probes expands, resulting in an

enhanced NBD signal. H2A4B4 LNP achieved significantly more membrane fusion than H2A4B3, implying that asymmetric branching chains improve the membrane fusion capacity of the LNP, particularly at longer incubation periods (Fig. 3 *C* and *D*). The relative ability of H2A4B4 and H2A4B3 in LNP-mediated membrane destabilization was further investigated using red blood cell (RBC) hemolysis assays. Erythrocytes are a model system for testing membrane destabilization due to the similarity in lipid bilayer structure between erythrocytes and endosomes (38). In this hemolysis model for membrane disruption, H2A4B4 LNP exhibited greater hemolysis compared to H2A4B3 LNP, affirming the enhanced rupture capabilities of the asymmetric branching chains (Fig. 3*E*). This result held true in low pH levels, mimicking the acidic endosomal compartment (Fig. 3*F*). A major “bottleneck” for LNP-mediated mRNA delivery is the successful escape of mRNA cargo from endosomes, so insight into the lipid structures that facilitate this process is of great interest for mRNA delivery applications.

To further improve the mRNA transfection potency of lipid H2A4B4, the head group was structurally modified by adjusting the number of secondary carbons, or methylene units, while maintaining the original dimethylamine structure of H2A4B4 without using diethylamine or pyrrolidine. Previous studies have shown that smaller head groups enable ionizable lipids to adopt a reverse nonbilayer structure (39). Additionally, the distance between the ester bond and the tertiary amine can affect the ionizability of the nitrogen atom, altering the pKa of the ionizable lipid. This change would be sufficient to affect the apparent pKa value of the LNP without significantly altering the molecular size of the hydrophilic region (40). Hence, the lipids A4B4-S3 and A4B4-S4, which share the same top-performing tail combination as H2A4B4, were synthesized for further evaluation (Fig. 3*G*). Surprisingly, the lipid A4B4-S3, which contains three methylene units, exhibited significantly higher mRNA transfection potency compared to the lipids H2A4B4 and A4B4-S4, which had two and four methylene units, respectively (Fig. 3 *H* and *I*). Although the spacers in the lipids differed only by the number of methylene units, their corresponding LNPs exhibited significantly different mRNA transfection potencies, indicating that the ionizable headgroup of the lipids H2A4B4 and A4B4-S4 are not optimal. Using the TNS [6-(*p*-Toluidino)-2-naphthalene sulfonic acid sodium salt] assay, we found that the slight change in spacer length strongly impacted the apparent pKa of the entire LNP (6.79 to 7.327). The pKa of ionizable lipids is a crucial parameter in optimizing the mRNA delivery performance of LNPs. In the development of their mRNA vaccine, Moderna synthesized 30 lipids and compared their efficacy to MC3 (41); their results showed that the optimal pKa range for I.M. delivery was 6.6 to 6.9, thus lipid SM-102 was successfully utilized in the COVID-19 vaccine “mRNA-1273” (42). Interestingly, our finding is that the A4B4-S3 LNP, which has an apparent pKa of 6.912, outperformed the H2A4B4 LNP, the pKa of which (6.792) falls more squarely within their range (Fig. 3 *J–L*). The pKa of reported ionizable lipids typically falls under 7.0, rendering them neutral under physiological conditions (pH = 7.4) (30). This neutrality facilitates the formation of stable complexes with mRNA. Nonetheless, in acidic pH environments such as endosomes (pH < 6.0), where protonation occurs, these lipids become positively charged, facilitating their dissociation from mRNA complexes and enhancing endosomal escape (40). These findings highlight the importance of ionizable lipid structure in LNP efficacy and demonstrate that the ability to fine-tune the pKa of ionizable lipids through spacer length variation is a powerful tool in designing new lipids.

The Top-Performing Lipid, A4B4-S3, Forms Greater Hydrogen Bonds with the mRNA Ribose-Phosphate Complex. Previous studies have demonstrated that altering the ability to form hydrogen bonds between lipid headgroups and RNA bases affects mRNA transfection potency (43). To better understand the impact of modified spacers, regions between the tertiary amine and ester bonds, on the potency of ionizable lipids in mediating mRNA transfection, molecular docking was performed to assess the interaction forces between the lipid headgroups and the mRNA ribose-phosphate complex (*SI Appendix, Table S2*). By docking different mRNA ribose-phosphate complexes with lipids A4B4-S3 and A4B4-S4, which contain three and four methylene units, respectively, the hydrogen bond interactions between the lipid headgroups and mRNA ribose-phosphate helix could be evaluated (Fig. 4*A*). The electrical distribution, as reflected by the calculated electronic static potential (ESP), indicates that the mRNA ribose-phosphate contains negatively charged regions near the phosphate oxygen atoms (red area) and positively charged regions near the RNA base nitrogen atoms (blue area). Meanwhile, the headgroups of the A4B4-S3 and A4B4-S4 lipids display different electrical distributions; both exhibit significant electronegativity in the ester-bonded region, but the A4B4-S3 headgroup is electroneutral while the A4B4-S4 headgroup contains additional regions of electronegativity (Fig. 4*B*). These results were consistent with the TNS assay detailed above. In addition, carboxyl oxygen atoms, which are commonly present in ionizable lipids, are considered the primary hydrogen bond acceptors. The results from the molecular docking experiments suggest that the interaction between the A4B4-S3 lipid and ribose-phosphate complexes, particularly with guanine (G) and cytosine (C), involves a larger number of hydrogen bonds compared to A4B4-S4 (Fig. 4*C*). Interestingly, previous studies have shown that the GC content in mRNAs is closely related to mRNA fate, correlating with both mRNA stability and translation efficiency (44). Given our results, it is possible that the GC content of mRNA may also affect the efficiency of LNP-mediated mRNA delivery.

To further investigate the structure-activity relationship of our top lipids for mRNA delivery, we theoretically analyzed the hydrolysis byproducts of our lipids and hypothesized how they may affect mRNA cargo release. Previous studies suggest that the hydrolysis of ester bonds within cells or tissues can impact the interaction between ionizable lipids and mRNA (23). The hydrolysis of the A4B4-S3 and A4B4-S4 lipid ester bonds yields a dimethyl amino dicarboxylic acid fragment and two relatively short and hydrophilic primary alcohols (*SI Appendix, Fig. S10*). The resulting negatively charged carboxylic acid could neutralize or reverse the positive charge associated with the amino headgroup, thereby enhancing the dissociation of the lipid–mRNA complex and facilitating mRNA escape from the endosome.

Delivery of Cre-Recombinase mRNA and Human EPO mRNA to Mice via the Top-Performing LNPs. Ionizable lipids with a neutral charge at physiological pH are designed to reduce toxicity while maintaining cellular uptake. We assessed the toxicity and immunogenicity of A4B4-S3 using WST-8 assays for cytotoxicity and luciferase dual reporter assays (NF- κ B-SEAP and IRF-Lucia) for immune activation. A4B4-S3 showed no significant immunogenicity or cytotoxicity compared to the commercial benchmark SM102, supporting its suitability as a safe, nonimmunostimulatory lipid for mRNA delivery (*SI Appendix, Fig. S11*). To assess the efficacy of our top-performing LNPs in mediating gene editing and cell tropism, genetically engineered tdTomato reporter Ai9 mice were utilized. The reporter Ai9 mice

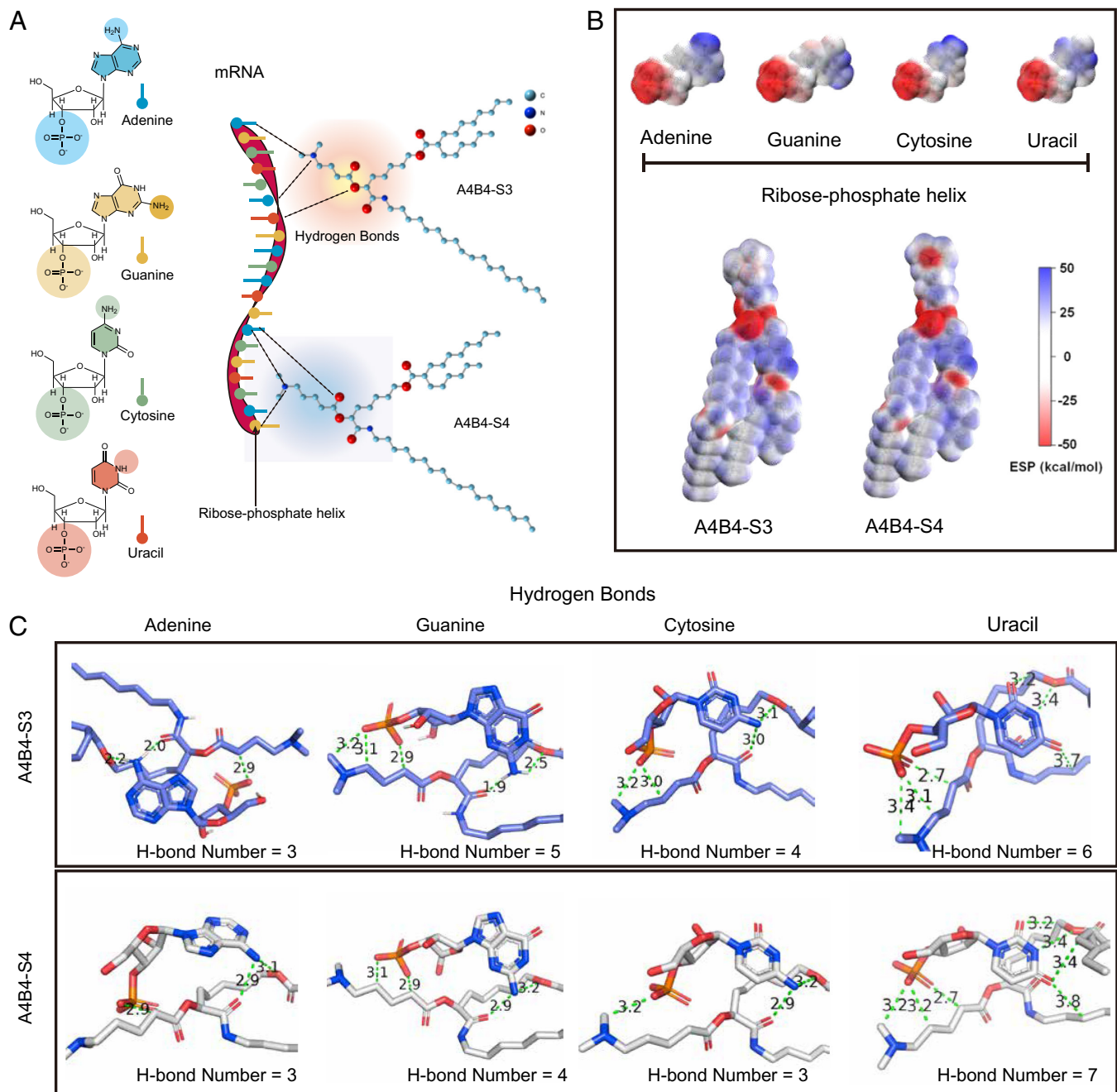


Fig. 4. Ionizable lipid-mediated mRNA delivery mechanism correlations with lipid headgroup. (A) Diagram depicting the interactions between ribose phosphate complexes and ionizable lipids (A4B4-S3 and A4B4-S4). (B) The electrostatic potential distributions of the ribose-phosphate complexes, lipids A4B4-S3 and A4B4-S4 are shown. Regions with negative electrostatic potential are indicated in red, while regions with positive electrostatic potential are shown in blue. (C) The schematic diagram illustrates the docking interactions between lipid molecules and ribose-phosphate complexes. The green dashed lines indicate the presence of hydrogen bonds.

allow visualization of gene editing mediated by Cre-recombinase mRNA (mCre). Once translated from mRNA, Cre-recombinase facilitates the excision of LoxP-flanked stop cassettes, resulting in the expression of a red fluorescent tdTomato protein within edited cells (Fig. 5A). Lipids A4B4-S3 and A4B4-S4 were formulated into LNPs along with mCre and compared with the SM-102 LNP after Retro-orbital injection. After two doses, tdTomato fluorescent signals in various organs of Ai9 reporter mice were visualized using IVIS imaging (SI Appendix, Fig. S12). Consistent with previous results, A4B4-S3 LNP achieved significantly higher tdTomato fluorescent signals than A4B4-S4 LNP. Additionally, A4B4-S3 LNP showed Cre-mediated gene editing efficiency comparable to SM-102 LNP. To further validate these findings, tissue sections were

analyzed using confocal microscopy to visualize tdTomato-positive cells (Fig. 5B). The cells edited by mCre were evenly distributed throughout the mouse liver following LNP administration. Notably, a remarkable increase in tdTomato-positive cells was observed after dosing with A4B4-S3 LNPs compared to A4B4-S4 and SM-102 LNPs (SI Appendix, Fig. S12). Flow cytometry was conducted to quantify the percentage of tdTomato-positive cells (Fig. 5C). Remarkably, A4B4-S3 LNPs exhibited superior performance to SM-102 LNPs in transfecting both hepatocytes and immune cells. A4B4-S3 LNPs exhibited superior gene-editing efficacy compared to SM-102 LNPs, achieving a higher percentage of gene-edited CD45⁺/CD31⁺ cells (56% vs. 48%) and overall immune cells (34% vs. 28%) in the liver. Additionally, A4B4-S3

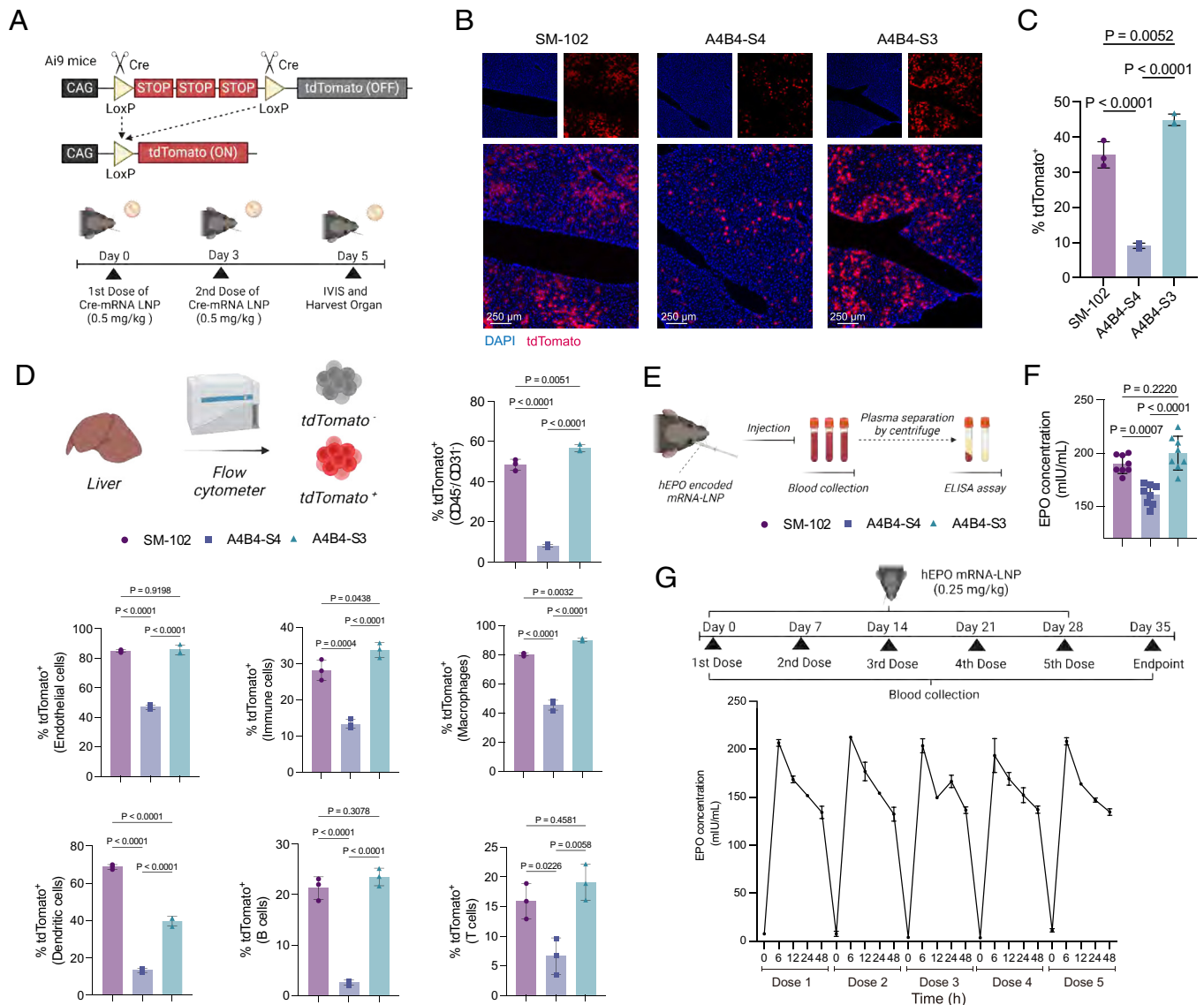


Fig. 5. In vivo delivery of mCre and hEPO mRNA to mice through retro-orbital administration. (A) Schematic illustration of Cre-mediated mRNA delivery in the Ai9 reporter mouse. Ai9 mice were retro-orbitally administered with mCre LNPs (0.5 mg kg⁻¹ mCre per mouse, n = 3). The livers were collected and analyzed by flow cytometry. (B) Representative fluorescence images of liver sections (Channel: tdTomato and DAPI, Scale bar: 250 μ m). (C) Histogram analysis for fluorescence images, quantified by ImageJ. (D) Quantification of tdTomato⁺ hepatocytes, endothelial, macrophages, dendritic cells, B cells, and T cells (n = 3). (E) Schematic illustration of retro-orbital injection of hEPO-containing LNPs. After 6 h dosing, blood was collected. Plasma was isolated for further ELISA. (F) Histogram analysis for hEPO concentration by ELISA (0.25 mg kg⁻¹ hEPO mRNA per mouse, n = 8). (G) The administration of A4B4-S3 hEPO LNPs was repeated every 7 d for a total of five doses, and plasma hEPO levels were measured by ELISA at different time points (0.25 mg kg⁻¹ hEPO mRNA per mouse, blood was collected at 0, 6, 12, 24, and 48 h). Statistical significance was assessed using a two-tailed Student's *t* test. **P*-value < 0.05, ***P*-value < 0.01, ****P*-value < 0.005. Data are presented as mean $\pm$ SEM.

LNPs also exhibited comparable editing efficacy with SM-102 LNPs in endothelial cells and specific immune cells (Fig. 5D and SI Appendix, Fig. S13).

Finally, the ability of the LNPs to deliver human erythropoietin (hEPO) mRNA, which encodes for a secreted therapeutic protein, was tested. hEPO protein expression was measured in blood plasma by ELISA; the result showed the highest concentration of hEPO protein after mRNA delivery by A4B4-S3 LNP, compared to A4B4-S4 and SM-102 LNPs (Fig. 5E and F). To investigate the potential use of the A4B4-S3 LNP for a repeat dosing regimen, retro-orbital administration of EPO mRNA-LNP was performed, as has been previously reported for other LNPs (45). In brief, mice were administered 0.25 mg kg⁻¹ of hEPO mRNA-LNP and evaluated for EPO protein expression over 48 h, followed by a 1-wk interval before repeating the administration for 5 wk. Subsequent observations revealed a consistent peak in plasma hEPO following

each administration (Fig. 5G). No significant adverse effects were observed in the mice throughout the duration of the experiment. Since our LNPs primarily accumulate in the liver, we assessed whether repeated dosing of A4B4-S3 LNPs would cause any adverse effects on hepatic tissue. Following repeated doses of hEPO mRNA-LNP injections, we measured the levels of alanine aminotransferase (ALT) and aspartate aminotransferase (AST) in the serum to assess the potential hepatic injury. Additionally, we conducted histopathological examinations of liver tissue using hematoxylin and eosin (H&E) staining to detect any morphological changes. Both biochemical and histological assessments revealed no significant abnormalities, indicating that A4B4-S3 did not induce hepatotoxicity, likely due to its biodegradable structure (SI Appendix, Fig. S14). Taken together, these datasets provide strong support for the safety and feasibility of the A4B4-S3 LNP for repeat dosing.

Conclusion

In conclusion, the chemical optimization of lipid structures for effective mRNA delivery remains a critical challenge in nucleic acid therapeutics. Despite extensive research into the diverse functions of lipid structures, identifying the optimal ionizable lipid largely depends on empirical knowledge or experimental iteration. Therefore, leveraging the modular regulation of P-3CR enables the rapid synthesis of lipid libraries that can be used to investigate the functional impact of subtle differences in chemical structure, and use these findings to rationally design improved lipids. Moreover, our findings indicate that the mRNA delivery efficiency of the H2A4B4 lipid, as identified in the batch test, can be significantly augmented through chemical modification employing modular chemistry. Utilizing molecular docking analysis, we demonstrated that variations in spacer length impact the number of hydrogen bonds formed between the ionizable lipid and the ribose phosphate complex and affect the electronic environment of the protonatable nitrogen in the head group. This study provides valuable insight into how the number of methylene units in the linker region can impact the complexation of ionizable lipids with nucleic acids, and overall mRNA transfection potency. The resulting ionizable lipid candidate A4B4-S3 successfully achieved potent *in vivo* Cre-mediated gene recombination and hEPO mRNA expression, superior to clinical benchmarks. Our study presents a robust combinatorial chemistry approach for the rapid synthesis and screening of ionizable lipids, facilitating rational optimization of candidates using modular chemistry. This work offers a strategy for the rational design and accelerated discovery of ionizable lipid candidates for mRNA delivery.

Materials and Methods

Extended materials and methods are available in [SI Appendix](#).

Lipid Library and Ionizable Lipid Synthesis. To synthesize ionizable lipids for the lipid library, we utilized the Passerini 3CR (P-3CR), which involves the reaction between carboxylic acid groups (-COOH), aldehyde groups (-CHO), and isocyanide groups (-NC) (6). In this method, equimolar amounts of carboxylic acid, aldehyde, and isocyanide (1:1:1 molar ratio) were dissolved in dichloromethane within a 96-well PCR plate with a sealing mat. The reaction mixture was stirred at room temperature for 24 h. For *in vitro* high-throughput screening, solvents were removed from the lipid mixtures using a vacuum-heated chamber, and the lipids were resuspended in ethanol (EtOH). When purification was required, the lipids were purified using a BUCHI flash chromatography system. Structural confirmation of the synthesized lipids was performed using proton NMR (^1H NMR) spectroscopy at 400 MHz and carbon NMR (^{13}C NMR) spectroscopy at 500 MHz (Bruker spectrometer), along with mass spectrometry using a Q Exactive HF-Orbitrap instrument (Thermo). Detailed synthetic procedures and characterization data are provided in [SI Appendix](#).

LNP Formulation. Except for the high-throughput screening and batch-testing stages, all other experiments utilized purified lipids. LNPs were formulated by combining an ethanol phase containing the lipids with an aqueous phase containing mRNA using a T-junction device connected to syringe pumps. The mixture was then dialyzed against $1 \times$ PBS in a 10,000 MWCO cassette (Thermo Fisher) at 4 °C for 12 h to exchange buffers. Specifically, the organic phase was prepared by dissolving the synthesized ionizable lipid mixture, helper phospholipid DSPC (Avanti), cholesterol (Avanti), and C14-PEG2000 (Avanti) in ethanol at molar ratios of 50:10:38.5:1.5, matching the composition of SM-102. The lipids MC3 and SM-102 were sourced from Echelon Bioscience. The aqueous phase consisted of mRNA dissolved in 10 mM citrate buffer at pH 4.0 (Fisher). The final LNP formulation achieved a total mRNA concentration of 0.1 $\mu\text{g}/\mu\text{L}$. The mRNA variants used in this study included mLuc (firefly luciferase mRNA from TriLink), mCre (Cre-recombinase mRNA modified with 5-methoxyuridine [5moU], from TriLink), and hEPO (human erythropoietin mRNA, 5moU, from TriLink).

LNP Characterization. The size and polydispersity index (PDI) of LNPs were evaluated using dynamic light scattering (DLS). LNP samples were diluted in phosphate-buffered saline (PBS, pH 7.4) to achieve a total mRNA concentration and transferred into cuvettes. Measurements were performed at 25 °C using a Malvern Nano ZS Zetasizer. Instrument settings included a measurement duration of 2 s per run with automatic determination of the number of runs and attenuation. Particle size was reported as the Z-average diameter.

To determine the apparent pKa of LNPs, a TNS binding assay was conducted. LNPs were diluted and mixed with TNS in buffered solutions to a final volume, resulting in an LNP concentration of 75 μM . Buffers ranging from pH 3 to 10 were prepared using boric acid, imidazole, sodium acetate, glycylglycine, and sodium chloride. Fluorescence measurements were taken using a Cytation 5 Cell Imaging Multi-Mode Reader (BioTek) with excitation at 321 nm and emission at 445 nm after the addition of TNS. The pH of each sample was measured afterward. Fluorescence data were fitted to the Henderson-Hasselbalch equation to calculate the pKa value.

mRNA concentration and encapsulation efficiency were analyzed using the RiboGreen assay, specifically utilizing the Quant-iT RiboGreen assay kit from Invitrogen. For the details of calculating the encapsulation efficiency by RiboGreen assay, samples were diluted in pH 8.0 Tris-EDTA (TE) buffer. The samples were then adjusted to a final concentration of 0.1 ng/ μL in the cuvettes. Fluorescence was measured using a Cytation 5 Cell Imaging Multi-Mode Reader (BioTek) with an excitation wavelength set at 500 nm and an emission wavelength at 525 nm. The standard curve was constructed by plotting fluorescence intensity against known mRNA concentrations, followed by a linear regression analysis to determine RNA concentrations in the LNP formulations. The encapsulation efficiency was determined by comparing the two measurements and applying the appropriate calculation:

$$\text{Encapsulation Efficiency (\%)} = \frac{\text{Total Fluorescence} - \text{Unencapsulated Fluorescence}}{\text{Total Fluorescence}} \times 100\%.$$

Cell Culture. To assess cytotoxicity, HEK-293 cells were seeded into 96-well plates at a suitable density (10^5 cells per well) and allowed to adhere overnight at 37 °C in a humidified atmosphere with 5% CO_2 . The following day, cells were transfected with LNPs and incubated for 24 h under the same conditions. After the incubation period, 10 μL of WST-8 reagent was added directly to each well containing 100 μL of culture medium. The cells were then incubated for an additional 1 h at 37 °C to allow for the formation of the formazan product.

After incubation with WST-8, the plate was shaken gently on a microplate shaker for 1 min to ensure homogeneous mixing. The absorbance of each well was measured at a wavelength of 450 nm using an Agilent BioTek Cytation 5 microplate reader. The absorbance at 450 nm correlates with the amount of formazan dye generated by cellular dehydrogenases in viable cells, providing a quantitative measure of cell viability.

To detect NF- κB and IRF activation in THP1-Dual Cells, the culture medium was collected 24 h after transfection. The medium was then mixed with QUANTI-Lu (InvivoGen) and QUANTI-Blue (InvivoGen) assay reagents, respectively, following the manufacturer's protocols to measure Lucia luciferase and SEAP levels. Relative luminescence units (RLU) were recorded using a luminometer, and absorbance at 655 nm was measured using an Agilent BioTek Cytation 5 microplate reader to quantify SEAP activity.

Animals Experiment. All experimental procedures are ethically approved, and all animal studies were approved and conducted in compliance with the University Health Network Animal Resources Centre guidelines. Female and male C57BL/6, BALB/C, and B6. Cg-Gt (ROSA)26Sor^{tm9(CAG-tdTomato)Hze/J} mice (4 to 8 wk) were obtained from the Jackson Laboratory.

In Vivo Batch Test. For *in vivo* high-throughput batch testing, a lipid-mRNA mixture within a classification group was combined and dialyzed, followed by i.m. injection into mice. Subsequently, 0.25 mg kg^{-1} of mLuc was utilized for the first batch-based LNP analysis, organized into 9 groups based on head groups, with each group comprising 16 LNPs mixture. For the second round of batch-based LNP analysis, 0.8 mg kg^{-1} of mLuc was administered to each mouse, organized into 4 groups based on tail A groups, with each group consisting of 4 LNPs mixture.

Additionally, 0.05 mg kg⁻¹ of mLuc was used for the third round of individual LNP analysis, injected for each lipid group. After 6 h postinjection, to measure in vivo bioluminescence, 0.2 mL of Xenolight D-luciferin (10 mg mL⁻¹ in Dulbecco's phosphate-buffered saline, PerkinElmer) was intraperitoneally injected into mice. Mice were anesthetized in an oxygen anesthesia chamber containing 2.5% isoflurane, and imaging was conducted using an in vivo imaging system (IVIS, PerkinElmer) 10 min after the administration of luciferin. The bioluminescence was quantified using Living Image software (PerkinElmer, version 4.8.0).

Extracellular/Intracellular FRET Assays. To create a simulated anionic liposome interior, we employed the following method: DOPS: DSPC: DOPE: NBD-PE and N-Rh-PE were combined in a molar ratio of 25:25:48:1:1. The FRET fluorescence probes NBD-PE and Rho-PE, which were conjugated with DOPE, were incorporated into the same simulated nanoscale particles. Simply put, they were initially mixed in chloroform, followed by rotary evaporation and vacuum drying for 2 h to form a thin lipid film. Subsequently, the dried film was hydrated in PBS (pH 7.4), and subjected to 20 min of ultrasound treatment, and the overall lipid concentration was maintained at 1 mM.

Fusion between lipids and simulated interior anionic liposomes was assessed through FRET measurements. The FRET with rhodamine caused a decrease in NBD fluorescence, and once lipid fusion occurred, the NBD signal increased due to the greater distance between the two probes. In the experiment, PBS (pH 5.5) was added to a black 96-well plate (100 μ L per well), and 1 μ L of simulated interior anionic liposomes (1 mM) was introduced into each well. Subsequently, 10 μ L of LNP or 1 μ L of lipid suspension was added to the wells. After incubating at 37 °C for 10, 20, and 30 min, fluorescence was measured on a microplate reader at Ex/Em = 465/520 nm (F). Wells containing only PBS with simulated interior anionic liposomes were designated as negative controls (F min), while wells containing probes incubated with Triton X-100 solution (2 wt.%) were used as positive controls (F max). The percentage of lipid fusion (%) was calculated as

$$\text{Lipid fusion (\%)} = \frac{(F - F_{\text{min}})}{(F_{\text{max}} - F_{\text{min}})} \times 100\%.$$

Hemolysis Test. Freshly collected whole blood was used to isolate mouse red blood cells (RBCs) through centrifugation at 1,000 g for 5 min. The isolated RBCs underwent five washes with PBS buffer (pH 7.4). Subsequently, the RBCs were suspended in PBS at pH 7.4 and 5.5. The lipid nanoparticles (LNPs) were formulated in vitro following the method outlined above, wherein lipids were dissolved in chloroform, rotary evaporated, and vacuum dried to form lipid films. PBS (pH 7.4) was added to the films, and the mixture was sonicated for 20 min to obtain a pellet suspension. A 96-well plate was used to combine the erythrocyte suspension and the calculated LNP. Following incubation at 37 °C for 1 h, the erythrocyte solution underwent centrifugation at 1,000 g for 5 min, and the supernatant containing hemoglobin was collected. Hemoglobin content was evaluated using an enzyme marker at 540 nm. Negative controls consisted of erythrocyte suspensions incubated in PBS, while positive controls involved incubation in Triton X-100 solution (1 wt.%). Cell viability assessments for both lipids and LNP were conducted under the same conditions, ensuring that hemolysis was not induced by cytotoxicity.

Theoretical Calculations. For theoretical calculations, the target molecules, including the ionizable lipids A4B4-S3 and A4B4-S4, as well as the four nucleobases adenine (A), guanine (G), cytosine (C), and uracil (U)—were initially drawn using ChemDraw 20.0. These molecular structures were exported as SD files containing three-dimensional atomic coordinates. The files were then imported into the Gaussian 16 software for computational analysis (46). Geometry optimizations were performed at the B3LYP-D3(BJ)/def2-SVP level of theory, and frequency calculations were conducted to confirm that the optimized structures correspond to true local minima on the potential energy surface. Following geometry optimization, single-point energy calculations were carried out to obtain accurate energies of the molecular systems. For the analysis of hydrogen-bonding interactions between the ionizable lipids and nucleobases, GaussView was utilized to construct hydrogen-bonded complexes comprising pairs of small molecules. The relative positions of the molecules were manually adjusted to establish reasonable hydrogen bonds, typically involving a hydrogen donor group (e.g., O-H or N-H) interacting with an electron-rich acceptor atom (e.g., O or N). The B3LYP-D3(BJ)/def2-SVP computational method and basis set were employed to calculate

the interaction energies of the hydrogen bonds, and PyMOL was used for the visualization of the complexes. Subsequently, electronic structure analyses were performed using Multiwfn version 3.8 (development version) (47). Formatted checkpoint files generated by Gaussian were utilized to compute molecular electrostatic potential (MEP) surfaces, providing insights into charge distribution and molecular reactivity. Finally, the results from Multiwfn were visualized using Visual Molecular Dynamics (VMD) version 1.9.3 to facilitate the interpretation and presentation of the data (48).

mRNA Delivery in the tdTomato Cre Reporter Mice Model. Cg-Gt(ROSA)26Sor^{tm9(CAG-tdTomato)Hze/J} miceCre reporter mice were retro-orbital injected with Cre mRNA encapsulated LNPs (0.5 mg/kg) through the tail vein at day 0, 3. Mice were killed on Day 5 for tissue collection. For flow cytometry of Ai9 mice, mouse livers were harvested, and liver cells were isolated and prepared for staining. Initially, liver tissue was finely chopped using a blade and placed in 1 mL of digestion medium consisting of collagenase type 4 (100 units/ μ L) and DNase I (50 units/ μ L). Subsequently, the tissue mixture was transferred to a 15 mL centrifuge tube containing an additional 4 mL of digestion medium and incubated at 37 °C for 1 h with shaking. Next, the liver solution was filtered using a 70 μ m filter and washed once with 1 $\times$ PBS. A cell pellet was obtained by centrifuging for 10 min at a speed of 500 g at 4 °C. The supernatant was removed, and the cell pellet was resuspended in 4 mL of 1 $\times$ RBC lysis buffer (BioLegend, 420301) and incubated at room temperature for 5 min. After incubation, 10 mL of PBS was added to stop red blood cell lysis. The solution was then centrifuged again at 500 g for 10 min to obtain a cell pellet. The single cells were resuspended in cell staining buffer (2% FBS containing 1 $\times$ PBS) to make a cell solution at a density of 5 $\times$ 10⁶ cells/mL. 100 μ L of cell solution was transferred to a new EP tube to incubate with anti-mouse CD45 antibody (FITC, BioLegend, 1:1,000), anti-mouse CD31 antibody (Alexa Fluor[®] 594, BioLegend, 1:1,000), anti-mouse F4/80 (PerCP/Cyanine5.5, BioLegend, 1:200), anti-mouse CD11c (PE/Cyanine7, BioLegend, 1:800), anti-mouse CD3 Antibody (Brilliant Violet 605[™], BioLegend, 1:50), anti-mouse/human CD45R/B220 Antibody (Brilliant Violet 421[™], BioLegend, 1:50) for 30 min at room temperature to stain CD45⁺CD31⁺ cells (CD45⁺/CD31⁺ cells represent a population of nonhematopoietic and nonendothelial cells in the liver. This population is predominantly enriched with hepatocytes but may also include other nonparenchymal cell types, such as hepatic stellate cells, fibroblasts, and potentially other mesenchymal or stromal cells.), endothelial cells (CD45⁺CD31⁺ cells), macrophages (F4/80⁺ cells), dendritic cells (CD11c⁺ cells), T cells (CD3⁺ cells), and B cells (B220⁺ cells) (SI Appendix, Table S3). The stained cells were washed twice with 1 mL of 1 $\times$ PBS, then resuspended in 500 μ L 1 $\times$ PBS for flow cytometry. Zombie NIR[™] kit (BioLegend, 423105) was used to discriminate live cells. The PBS-only treatment group (NC) was used as the negative control. The data of flow cytometry were analyzed by FlowJo software.

For the confocal fluorescence imaging, all collected tissues were fixed in 4% buffered paraformaldehyde overnight at 4 °C, then dehydrated in 30% sucrose overnight at 4 °C. Next, a cryosection was performed on tissues after being frozen at optimum cutting temperature. Cell nuclei were stained with DAPI (1:1,000). Zeiss LSM700 was used to obtain and process images.

In Vivo Therapeutic hEPO mRNA Delivery. SM-102, A4B4-S3, and A4B4-S4 LNPs were formulated with hEPO mRNA into LNP. To assess the kinetics of the hEPO protein, approximately 100 μ L of blood was drawn from the facial vein of each mouse and transferred into EDTA-coated tubes. The blood samples were collected from different mice at each time point within the same treatment group for serum analysis. The collected blood samples were then centrifuged at 2,000 g for 20 min. The resulting plasma, the upper layer, was separated and used for ELISA testing. Plasma samples were aliquoted and stored at -80 °C until they were analyzed. hEPO levels in mice were measured via the hEPO ELISA Kit (Abcam, Ab274397) according to the manufacturer's instructions. For repeated-dose experiments, hEPO mRNA LNPs were injected into mice via retro-orbital following the method outlined above (0.25 mg/kg, per mouse). Blood samples were collected from two mice at each time point (n = 2). Across the 6, 12, 24, and 48-h time points, a total of 8 mice were used in the same treatment group, with two mice allocated to each time point for consistent representation. AST and ALT levels were measured using a standard colorimetric assay kit following the manufacturer's instructions to assess liver function and potential hepatotoxicity. For histological analysis, liver tissues were harvested,

fixed in 10% neutral-buffered formalin, and embedded in paraffin. Sections of 4 to 5 μm thickness were cut from the paraffin blocks and stained with H&E for microscopic evaluation.

Statistical Analysis. The data were subjected to statistical analyses using GraphPad Prism 9 (GraphPad Software). A two-tailed unpaired Student's *t* test was conducted to assess the significance of the comparisons as indicated. Data are expressed as mean $\pm$ SD. *P* values <0.05 (*), $P < 0.01$ (**), $P < 0.001$ (***), and $P < 0.0001$ (****) were statistically significant.

Data, Materials, and Software Availability. All study data are included in the article and/or [SI Appendix](#).

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